

Software Test Case Report

Web Based Shopping Mall

Course	SE2030 Software Engineering
Group ID	Group 07
Topic	Web Based Shopping Mall
Coverage	Six assigned major functions; 10 individual test cases per member

This report presents functional test cases for the six core functions of the group project. Each member is identified with their assigned function and student registration number. Each test case states setup conditions, action steps, and the expected outcome.

Group Members and Work Allocation

Member	Registration Number	Assigned Function
Neluvinda T.M.D.N.	IT25101109	User and Stakeholder Management
Bandara N.V.	IT25101100	Product Catalog Management
Vaas W.P.S.D.	IT25101123	Shopping Cart Handling
Maliduwa M.G.B.B.	IT25101122	Order and Payment Handling
Weerawickrama K.K.	IT25101108	Product Feedback and Reputation Analytics
Ariyasinghe P.A.D.A.J.	IT25101088	Customer Support and Inquiry System CRUD

Test case format: Test case ID, scenario, preconditions/test data, test steps, expected result, and priority. Priority is High for core user journeys and access-control or data-integrity checks; Medium for supporting views and validation cases.

User and Stakeholder Management

Member: Neluvinda T.M.D.N. | Registration No.: IT25101109 | Test cases: 10

ID / Priority	Scenario and Preconditions / Test Data	Test Steps	Expected Result
US-01 High	Register customer with valid details Visitor is logged out; email is unused.	Submit name, email, password and customer role.	Customer account is saved; confirmation/success message is shown.
US-02 High	Reject customer registration with existing email An account already uses customer@example.com.	Submit registration using the same email.	Registration is rejected; no duplicate account is created.
US-03 Medium	Reject invalid registration details Visitor is on the registration form.	Submit malformed email or missing required fields.	Field validation errors appear; account is not created.
US-04 High	Authenticate with valid customer credentials Customer account is active.	Enter correct email and password on login.	Customer is authenticated and redirected to an allowed customer page.
US-05 Medium	Reject incorrect password Customer account exists.	Enter correct email and incorrect password.	Login is denied and no authenticated session is created.
US-06 Medium	Deny customer-only page to logged-out visitor No active session.	Open a customer-protected URL directly.	User is redirected to login or access is denied.
US-07 Medium	Deny role-restricted action to wrong role Logged in as customer; merchant management URL is known.	Request the merchant-only URL directly.	Access is denied; no merchant data is changed.
US-08 High	Submit merchant registration Visitor is logged out and merchant registration is available.	Submit valid merchant and shop profile details.	Merchant account is created in pending state for employee review.
US-09 High	Approve pending merchant as employee Logged in as platform employee; pending merchant exists.	Open pending merchant and choose approve.	Merchant status becomes approved and the result is shown.
US-10 High	Reject unauthorized merchant approval Logged in as customer or merchant; pending merchant exists.	Attempt approval URL/action.	Request is denied; merchant remains pending.

Product Catalog Management

Member: Bandara N.V. | Registration No.: IT25101100 | Test cases: 10

ID / Priority	Scenario and Preconditions / Test Data	Test Steps	Expected Result
PC-01 High	View unified product catalog At least two active merchants have listed products.	Open catalog page.	Available products from multiple merchants are listed.
PC-02 High	Search by matching product keyword Catalog contains a product named "Wireless Mouse".	Search for "mouse".	Matching product appears in results.
PC-03 Medium	Search with no matching keyword Catalog contains products but none match the term.	Search for a unique nonexistent term.	Empty-results message/state appears without an error.
PC-04 High	Filter catalog by category Products exist in two or more categories.	Select one category filter.	Only products in the selected category are shown.
PC-05 Medium	Filter catalog by price range Products exist below, within and above a range.	Set minimum and maximum price.	Only products whose prices fall within the range are shown.
PC-06 Medium	Filter catalog by merchant Two merchants have listed products.	Choose one merchant filter.	Results contain products from the selected merchant only.
PC-07 Medium	View product details A listed product exists.	Open its product detail page.	Description, price, stock and merchant information are displayed.
PC-08 High	Merchant creates product with valid data Logged in as approved merchant.	Submit name, category, price, stock and description.	Product is saved and appears in the catalog.
PC-09 High	Reject invalid product data Logged in as approved merchant.	Submit product with negative price or invalid required fields.	Validation errors appear; invalid product is not saved.
PC-10 High	Prevent merchant editing another merchant's product Two merchants and a product owned by the other merchant exist.	Attempt to open/update that product by its ID.	Access is denied; original product data remains unchanged.

Shopping Cart Handling

Member: Vaas W.P.S.D. | Registration No.: IT25101123 | Test cases: 10

ID / Priority	Scenario and Preconditions / Test Data	Test Steps	Expected Result
SC-01 High	Add a product to an empty cart Customer is logged in; product is available.	Add one unit of the product.	Cart contains the product with quantity one.
SC-02 High	Add same product again Cart already contains one unit of the product.	Add the same product again.	Existing cart line quantity increases; duplicate line is not created.
SC-03 Medium	View cart total Cart contains products with known prices and quantities.	Open cart page.	Each item subtotal and total equal current price multiplied by quantity.
SC-04 High	Update cart quantity to a valid positive value Cart contains an item.	Change quantity to a positive integer and submit.	Cart displays the updated quantity and recalculated total.
SC-05 Medium	Remove item by setting quantity to zero Cart contains an item.	Update its quantity to zero.	Item is removed from the cart.
SC-06 Medium	Remove item from cart Cart contains at least one item.	Choose remove for one item.	Selected item disappears; other items remain.
SC-07 Medium	Reject invalid negative quantity Cart contains an item.	Submit quantity less than zero.	Update is rejected and the cart retains a valid quantity.
SC-08 High	Keep cart associated with customer across sessions Customer has saved cart items.	Log out, then log in again as the same customer.	Previously saved items remain in the customer's cart.
SC-09 High	Prevent viewing another customer's cart item Two customers have separate carts; logged in as first customer.	Request update/remove for second customer's item ID.	Access is denied and the other cart is unchanged.
SC-10 High	Handle empty cart display Customer has no cart items.	Open cart page.	Empty-cart state is shown and total is zero.

Order and Payment Handling

Member: Maliduwa M.G.B.B. | Registration No.: IT25101122 | Test cases: 10

ID / Priority	Scenario and Preconditions / Test Data	Test Steps	Expected Result
OP-01 High	Complete checkout with available stock Customer cart has valid items and sufficient stock.	Submit checkout using the simulated payment flow.	Order and order items are created; cart is cleared; confirmation is shown.
OP-02 High	Reject checkout when stock is insufficient Cart quantity exceeds current product stock.	Submit checkout.	Checkout is blocked; no order is placed and stock is not deducted.
OP-03 Medium	Reject checkout for empty cart Customer cart is empty.	Open/submit checkout.	No order is created; user sees an appropriate empty-cart message.
OP-04 High	Record successful simulated payment Customer has a valid nonempty cart.	Select/confirm the successful simulated payment option.	Payment outcome is recorded as successful with the order.
OP-05 Medium	Handle failed simulated payment Customer has a valid nonempty cart.	Choose or trigger simulated payment failure.	Failure is reported and no paid/confirmed order is recorded.
OP-06 Medium	View customer order history Customer has placed prior orders.	Open order history.	Only the logged-in customer's orders are listed.
OP-07 Medium	View order summary Customer owns an existing order.	Open order details.	Items, quantities, prices, totals and current status are displayed.
OP-08 High	Merchant views own order items Merchant has products in customer orders.	Open merchant order management.	Merchant sees order items for their own products.
OP-09 High	Prevent merchant updating another merchant's item status Order contains items from two merchants.	Merchant A attempts to update Merchant B's item.	Update is denied; other merchant's item status is unchanged.
OP-10 High	Update own order item status Merchant owns an order item in pending state.	Change the item status to shipped/delivered using an allowed transition.	Owned item status updates and order rollup reflects item statuses.

Product Feedback and Reputation Analytics

Member: Weerawickrama K.K. | Registration No.: IT25101108 | Test cases: 10

ID / Priority	Scenario and Preconditions / Test Data	Test Steps	Expected Result
FR-01 High	Submit review for delivered purchase Customer owns a delivered order item not previously reviewed.	Submit a rating from 1 to 5 and review text.	Review is saved and appears for the product.
FR-02 High	Reject review for non-delivered purchase Customer owns an order item that is not delivered.	Attempt to submit a review for it.	Review is rejected and none is saved.
FR-03 Medium	Reject review for another customer's order item A delivered item belongs to another customer.	Submit review using that item ID while logged in as a different customer.	Access is denied; no review is saved.
FR-04 High	Prevent duplicate review for same purchased item Customer has already reviewed the delivered order item.	Submit another review for that same item.	Duplicate is rejected; original review remains.
FR-05 Medium	Validate rating range Customer has a reviewable delivered item.	Submit rating 0 or 6.	Validation rejects rating; review is not saved.
FR-06 Medium	Display full rating distribution Product has reviews with known ratings.	Open product details/review summary.	Counts for all five rating values are displayed, including zero counts.
FR-07 Medium	Return zero reputation for merchant with no reviews Merchant has no reviewed products.	Open reputation score/dashboard.	Score displays as 0.0 without an error.
FR-08 High	Flag product below rating threshold Product's review average is below 2.5 after a new review.	Submit the review that leaves average below threshold.	Product is marked for employee review.
FR-09 High	Clear flag after average recovers Flagged product receives enough good reviews to reach at least 2.5.	Submit a qualifying review.	Flag is recomputed and cleared.
FR-10 High	Merchant responds only to own review once Review belongs to merchant's product and has no response.	Submit a response; then attempt a second response.	First response is saved; second attempt is rejected.

Customer Support and Inquiry System CRUD

Member: Ariyasinghe P.A.D.A.J. | Registration No.: IT25101088 | Test cases: 10

ID / Priority	Scenario and Preconditions / Test Data	Test Steps	Expected Result
SI-01 High	Create inquiry linked to an order Customer is logged in and owns an existing order.	Submit inquiry with order reference and message.	Inquiry is created with OPEN status and linked to the order.
SI-02 High	Create inquiry linked to a product Customer is logged in and product exists.	Submit inquiry with product reference and message.	Inquiry is created with OPEN status and linked to the product.
SI-03 Medium	Reject inquiry with no order or product reference Customer is logged in.	Submit inquiry with neither related order nor product.	Validation rejects the request; no inquiry is saved.
SI-04 High	Read customer's submitted inquiries Customer has multiple inquiries.	Open customer inquiry list.	Customer's inquiries and their current statuses are displayed.
SI-05 Medium	Read inquiry details Customer owns an inquiry with at least one response.	Open the inquiry detail view.	Inquiry content, linked record, status and responses are shown.
SI-06 Medium	Prevent customer viewing another customer's inquiry Another customer owns an inquiry.	Request its detail page by ID.	Access is denied; private inquiry details are not shown.
SI-07 Medium	Update inquiry status as platform employee Platform employee is logged in; inquiry is OPEN.	Use status action to set IN_PROGRESS.	Inquiry status changes and the updated value is displayed.
SI-08 High	Resolve inquiry Platform employee is logged in; inquiry is IN_PROGRESS.	Choose resolve action.	Inquiry status becomes RESOLVED.
SI-09 High	Reject inquiry status change from customer Customer is logged in and owns an inquiry.	Attempt employee status update action.	Access is denied; inquiry status remains unchanged.
SI-10 High	Read inquiry list as platform employee Platform employee is logged in; inquiries exist.	Open employee inquiry list.	Employee can view submitted inquiries and their statuses.